

Statistical analysis of hospital data and Length of Stay Prediction via Machine Learning

Programming for Management Studies (MGTHN0144):

- Final Project -

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1 Introduction and problem statement

For our final project, we have decided to do an in-depth analysis of the dataset to be found under <https://www.kaggle.com/datasets/nehaprabhavalkar/av-healthcare-analytics-ii?resource=download>. The dataset deals with patient data from different hospitals (*see next section for details*). We have decided to focus on answering two managerial aspects of healthcare data science for our project, which are:

- (I) Statistical analysis of the data to find the department that produces the longest lengths of stay (LOS);
- (II) Prediction of the approximate length of stay of a new patient via a self-trained Machine Learning model.

2 Explanation of dataset and data cleaning

Our chosen dataset consists of 18 columns: case_id (Numeric), Hospital_code (Numeric), Hospital_type_code (Categorical), City_Code_Hospital (Numeric), Hospital_region_code (Categorical), Available Extra Rooms in Hospital (Numeric), Department (Categorical), Ward_Type (Categorical), Ward_Facility_Code (Categorical), Bed Grade (Numeric), patientid (Numeric), City_Code_Patient (Numeric), Type of Admission (Categorical), Severity of Illness (Categorical), Visitors with Patient (Numeric), Age (Categorical), Admission_Deposit (Numeric).

For cleaning the data, we have decided to implement the following algorithmic solution:

(i) loading the given dataset $\Rightarrow$ (ii) rows that miss at least one value are dropped $\Rightarrow$ (iii) overwriting the .csv-file with the new file $\Rightarrow$ (iv) changing the case_id s.t. it is continuous again after deleting rows (ensure consistant enumeration of the rows)

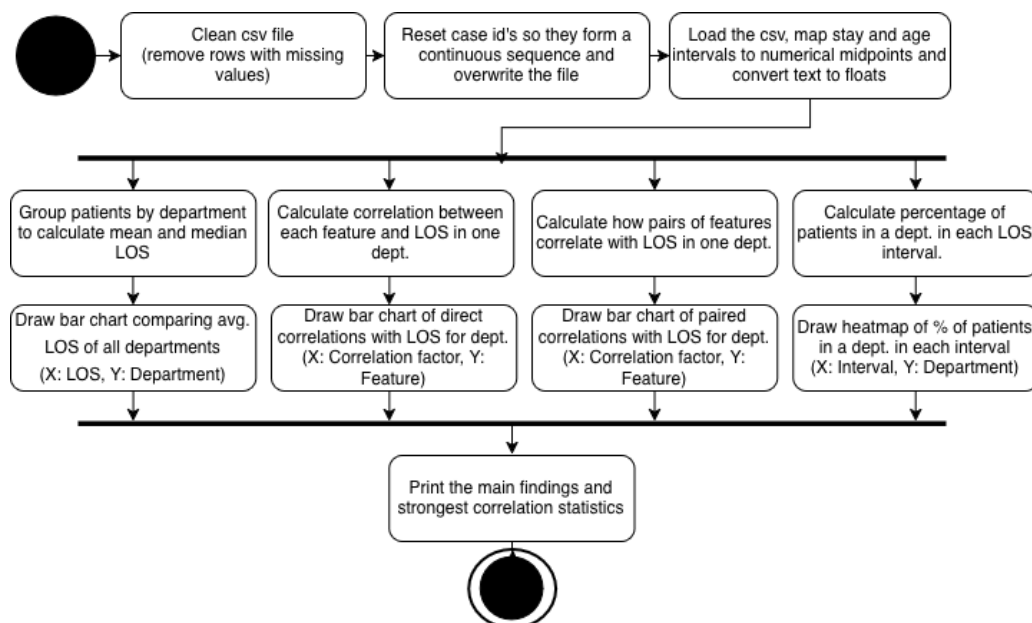
This reduced the number of rows (and thus datasets) from 455,495 to 313,793 rows (by 141,702 rows (=31.11%)), which were the basis for our further analysis.

3 (I) Which department produces the longest LOS and is thus the most costly for a hospital?

3.1 Abstract

Hospital financing in German healthcare is based on the G-DRG system. A financing timeframe is assigned upfront for different types of diagnoses, for which the expenses the patient causes the hospital are fully financed. However, for hospitalizations exceeding the predetermined threshold („obere Grenzverweildauer“), a hospital receives only a flat per diem payment! Further, since long hospital stays are usually tied to severe cases, patients often require specialized care and thus need a greater share of the available resources. It is therefore possible to conclude that an increased LOS of a patient increases the financial burden of a hospital. A privately-run hospital can identify the specific departments dealing with prolonged stays most often and then allocate resources to improve a department's efficiency, e.g. by taking investments in better and newer equipment, increasing human resources or hiring specialists. The goal of the following analysis is to, by using the provided data, find which department, on average, struggles with long Lengths of Stay (LOS) across hospitals and would therefore be the best candidate for a capital investment.

3.2 Process/algorithm of our solution (UML)

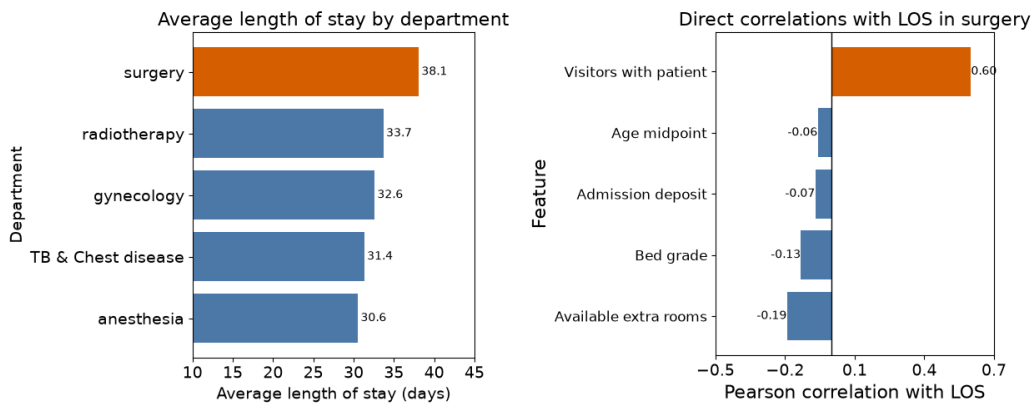


(Process of statistical analysis)

3.3 Statistical findings

We found that across hospitals included in the dataset, the surgical department has the longest average LOS with 38.1 days (median 35.5). Inside surgery, the strongest direct correlation with LOS is Visitors with patients ($r = 0.60$) However, it is important to note that the surgical department is the least to be found in the data (1,143 patients against e.g. 245,850 in gynecology). [Please also note that we

conducted further analysis, like computing second-order correlations, which we have not included in this report due to the given page limit. These are to be found in the submitted notebook.]



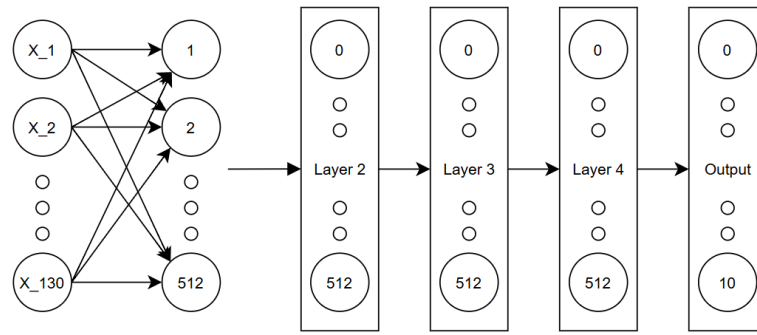
(Comparison of LOS by dept. and corr. with LOS within the surgical dept.)

3.4 Managerial advice

The surgical dept. exhibits the longest average LOS of patients and is thus the prime candidate for efficiency investments. It is to consider that since it produces the least amount of cases, it struggles most per patient, but limited absolute volume of costs. This is obviously still crucial when reducing economic losses under DRG system, as they accumulate mostly when exceeding the predetermined and fully financed LOS. Within the surgical departments, the number of visitors with the patient shows by far the strongest correlation with LOS, thus indicating case severity and care intensity. It is observable early in the stay and without further analysis, and could thus serve as early-warning indicator for cases which risk exceeding the DRG time window ("obere Grenzverweildauer"). **Recommended actions: increase direct investment into the surgical department and use a high visitor count during the stay as a possible indication for exceeding the allocated time window.**

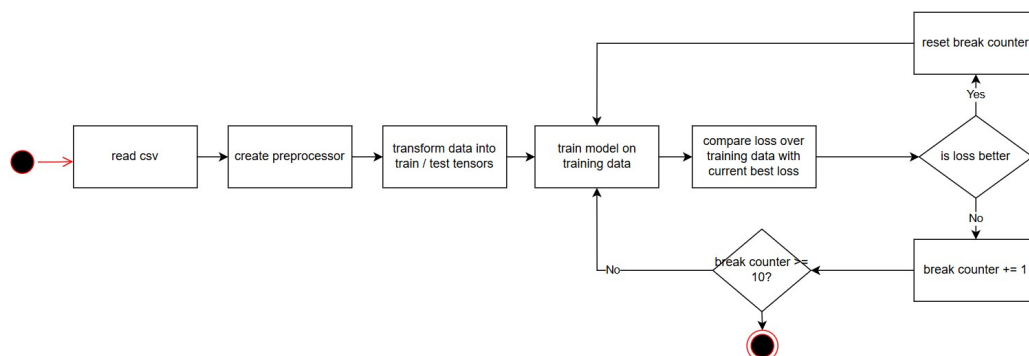
4 (II) ML Model

We trained a machine learning model to predict the LOS of a patient. We created a class to transform the raw data frame into usable train and test tensors. It contains a function that first splits rows into a training and test part and then normalizes and one-hot codes the given feature columns. After creating the label column by extracting the "stay" column, we transformed the data frame into tensors (X_{train} , X_{test} , y_{train} , y_{test}). For the prediction part, we created a neural network which is able to classify input data into the given stay groups. The model consists of one input layer with 130 input neurons, 1 hidden layer with 512 neurons each, and an output layer with 11 neurons that represent the 11 stay groups. After the input neuron and each hidden neuron, we applied the ReLU activation function. As an optimizer we chose standard Adam, and as the loss we decided to use CrossEntropyLoss.



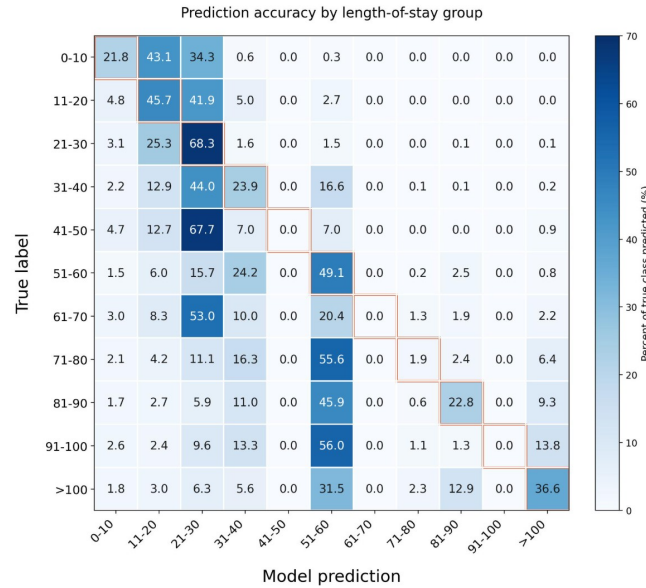
(Model architecture)

For the training loop we tracked the best test loss over the training process and stopped the training when no better test loss was achieved. We decided to use the test loss for the model performance evaluation because by tracking the performance of the model based on the already seen data (training data) we would have risked taking an overfitted model as the best one. After the training loop ended, we loaded the parameters that were saved in the best run of the model and then save this model in an external file. This allows us to use this pre-trained model to predict a dataset without having to train again.



(Training loop)

To check the real accuracy of our model and to see how the model performs on the different LOS groups, we split our data into 11 different sets where each consists only of rows that have same LOS. Then we predicted each class and tracked the prediction of the model (result below). E.g., the model predicts stay 0-10 in 21.8% of all cases as stay 0-10 (correct) but also predicts group 0-10 in 43.1% of all cases as 11-20. Some classes, such as 41-50, are predicted with 0% accuracy due to few rows of training data available for that class. During training, the model optimized for overall loss across all samples, so when the class is underrepresented, the model learns that it's statistically safer to predict a majority class instead of risking a wrong guess on a rare one. E.g., since 21-30 has by far the most training examples, the model learns a high prior probability for it and tends to default to predicting 21-30 even for inputs that belong to a minority class like 41-50. As a result, the model "gives up" on the rare classes in favor of minimizing loss on the classes it sees most often.



(Matrix: predicted LOS for each age group)

5 Final conclusion and managerial advise

We analyzed 313,793 cleaned hospital records to solve the two aspects mentioned. The following can be concluded:

1. Surgery is the department with the longest average LOS (38.1 days, median 35.5), ahead of e.g. radiotherapy (33.7) and gynecology (32.6). Within surgery, visitor count correlates most strongly with LOS ($r = 0.60$), making it a cheap early-warning indicator for stays at risk of exceeding the DRG time window. Surgical department is thus the prime candidate for targeted efficiency investment, though high-volume gynecology (245,850 cases) carries the largest absolute exposure.
2. A self-trained machine learning model predicts a newly admitted patient's LOS category from admission-time features with 42.3% test accuracy across 11 classes (vs. $\sim 9\%$ random baseline), with most errors falling into adjacent categories. Thus, it is accurate enough to support bed capacity planning, early discharge management, and staffing decisions at admission.

6 Roles and Contribution

Muhammad Sheheryaar Ghayas (03801683): Creation of UML diagrams, explorative analysis of data, creation of further visualisations and functions; **Vsevolods Mitrevics (03815799):** Creating functions for data loading, cleaning, aggregating and visualizing (for Task I); **Constantin Demuth (03818527):** Creating, training, testing and documenting ML model (for Task II); **Stanislaw Pohl (03822985):** Data analysis and exploration regarding suitable management questions, visualisations, research on healthcare system and contextual information; **Jannis Nickel (03813516):** Writing report, creating presentation slides, explorative Data analysis, Github Repo Management, Compilation of final notebook & deliverables.